CNAN

⇒ Fully Connected 3 layer Network.

1.) Number of MAC per layers
= No. of neurons in the layer × No of inputs for each neuron in the layer

Layer 1 = 784 × 256 = 2,00,704 MAC's
Layer 2 = 256 × 128 = 32,768 MAC's
Layer 3 = 128 × 10 = 1,280 MAC's.

So, these are the no. of MAC's for each layers.

2.) Total MAC's for one forward pass
= Sum of MAC's in all layers = Layer 1 + Layer 2 + Layer 3
= 2,00,704 + 32,768 + 1,280
= 2,34,752.

So, they are total 2,34,752 MAC's.

3.) Total no. of trainable parameters (Only weights). = Sum of all layers (Inputs × outputs)
= 2,34,752 weights

So, they are total 2,34,752 weights trainable parameters.

4.) Total weight memory in Bytes
(FP32 4 Bytes each)

= Total parameters × ~~registe~~ Bytes/weight.

= 234,752 × 4

= 939,008 bytes

So, they are total 9,39,008 weight memory in Bytes.

5.) Total Activation Memory

= Total no. of neurons in network × 4

= (784 + 256 + 128 + 10) × 4

= 4,712 Bytes.

So, they are total 4,712 Bytes Activation Memory.

6.) Arithmetic Intensity

$$= \frac{2 \times \text{Total MAC's}}{(\text{Weight Bytes} + \text{Activation Bytes})}$$

$$= \frac{2 \times 234752}{(9,39,008 + 4712)}$$

$$= \underline{0.4975 \text{ Flop/byte}}$$